EXPNO:4 DATE:	Launching Web Applications using App Engine Tools

Aim:

To launch and run a web application locally using Google App Engine tools (Flask) and verify it in a browser.

Pre-Requisites:

- Windows 10 Host Machine
- Oracle VM VirtualBox (Open Source)
- Ubuntu Desktop ISO image (Open Source)

Procedure:

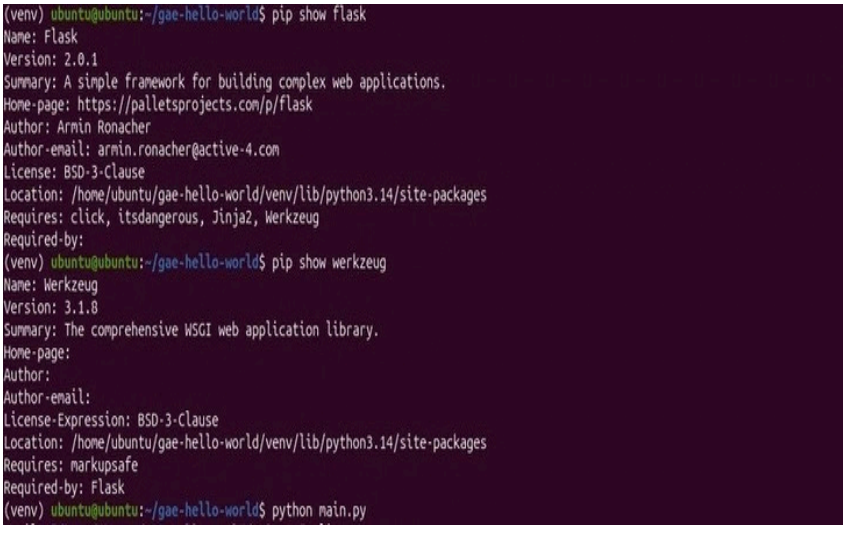
Step1:Upgrade Flask and Werkzeug

```
pipinstall--upgradeflaskwerkzeug
```

This upgrades Flask and its dependencies (Werkzeug, blinker, etc.) in the virtual environment.

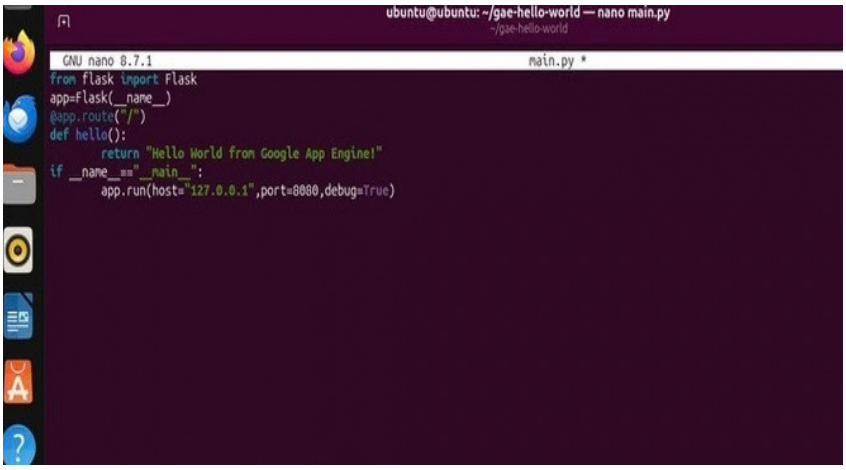
Step 2: Verify Installed Package Versions

```
pip show flask
pip show werkzeug
```



Step 3: Review Application Code (main.py)

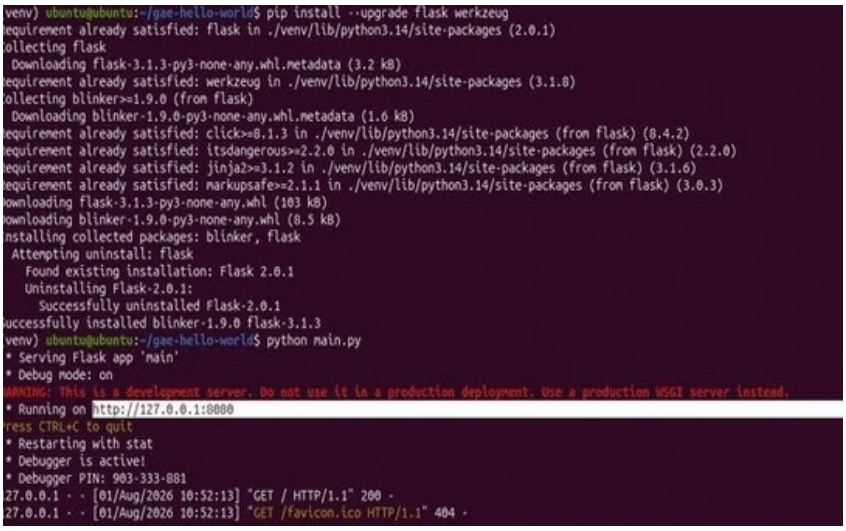
main.py defines a simple Flask route that returns a Hello World message:



Step 4: Run the Application

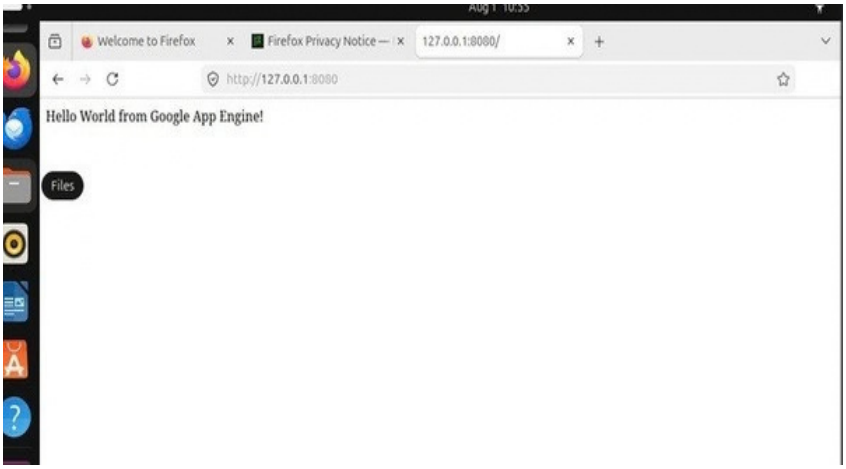
`python main.py`

Flask starts the development server on http://127.0.0.1:8080 with debug mode enabled.



Output:

The application was accessed from the browser at 127.0.0.1:8080, confirming successful deployment:



Result:

Thus, the Flask-based web application was successfully launched and tested locally using Google App Engine tools.